An agents-based tool for simulating farmers operation in irrigation districts

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ABSTRACT

Efficient management of water resources is increasingly important due to the expected scarcity caused by extended droughts in the close future. Often water resources are assumed to be managed by the water supplier from a complete knowledge centralized view using just hydrological criteria, but models that explicitly represent water consumers and the complex decision making and interactions that they take are required since they have conflicting goals, private knowledge and autonomous nature. In the context of a network of farmers, an entity delivering water (pumping station) computes the head pressure provided to the connected farmers, while farmers autonomously decide whether and when they connect themselves to the network. In this paper we designed and proposed an agent based system that includes the roles for farmers, a deliverer agent and a weather station providing the daily forecast. It also includes the protocols to rule out the cyclic interaction between them and an ontology to express the predicates, actions and concepts corresponding to the domain. Since decisions taken by farmers may carry on economic and legal consequences, such decisions are ruled by a BDI logic that provides explainability and full controllability. The agent system was implemented with GAMA platform to facilitate its integration with GIS topographic information of the network. The complete code is available in an open repository to be freely used by the interested community.

1. Introduction and State of the Art

Farmers in irrigated lands often group into irrigation districts where a set of common hydraulic infrastructures are shared to provide the farmers joining the irrigation district with water to cover the crop water necessities. Irrigation districts are frequently controlled by a central manager that manages the network considering the farmers requirements, the limitations from the infrastructure capacity and other external constraints. The operation usually follows the so-called on demand irrigation what implies that farmers receive the water upon request. In real networks, despite that idealized ambition, some limitations (i.e. those derived from the electric prices, the performance of the pumping station or the risks of network´s overload) hinder the strict on-demand immediate supply from the network.

Many uncontrolled factors (weather evolution, cropping strategies, crop evolution over the cropping season, soil properties, specific type of irrigation emitters, etc.) determine the farmer´s decisions on irrigation. In irrigation districts, farmers´ decisions determine the network operation conditions, the central manager decisions and, as a result of both, the quality of the supply to network offers to farmers. It means the amount, location and characteristics of the farmers connected to the network determine the network´s supply conditions (aggregated discharged flow and upstream energy head) and the flow rate and pressure head delivered at each hydrant.

The previous frame represents a scenario where different individuals interact with limited information, joining a common system to receive a good with a certain requirements and whose decisions have an impact on both the system and the remaining individuals. From a mathematical point of view, the most efficient solution for the operation of such a system would be achieved if the central manager has complete information and directly takes the farmers decisions. However, some constraints prevent from reaching that solution in the case of irrigation networks:

- Private knowledge what means farmers do not share complete information on the crop status, soil moisture or the expected time until the next connection to the network to irrigate.
- Autonomy of decision: farmers may not want to delegate the decision to connect to the network.

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- Conflict of interests among farmers since the network capacity is limited and the larger amount of farmers connected the greater risk of receiving poor performance.

The previous considerations suggest a distributed social decision making scenario is more realistic than the centralized operation. The features of this problem (dynamic and uncertain environment, privacy requirement, incomplete knowledge, different resources and abilities and conflicting goals) fit into the definition of a remote distributed management problem which is the scope of agents theory. Simulation through agents is a powerful mean to achieve automation in any human domain since agents represent autonomous and heterogeneous actors that simulate the complexity of human interactions in which their behavior is influenced by the unpredictable decisions and interactions with peers. Agents represent an abstraction of an autonomous and pro-active computational entity that interacts with their peers and a shared dynamic environment [1]. For instance, [2] suggests that agents may handle the complexity of coupling human and natural systems, facilitating the representation and understanding of the underlying dynamics. Agents theory has been previously used for simulating some topics related to agricultural water management (see for example [3], [4], [5] or [6]) and to address some questions tangentially linked to irrigation districts ([7], [8]) but to the authors´ knowledge no previous work uses the agents theory for simulating the operation of irrigation districts.

Seeking to leverage the suitability and performance of the proposed tools for the problem we are addressing, we present in this manuscript a theoretical framework for using agents to simulate the operation and performance of an irrigation district. Secondly, the evidence the method ability, we used the developed framework as a simulation tool in two case studies. To facilitate a broad use of the simulation tool we have provided all the source code freely in an open repository with the expectation of promoting its application in as many possible hypothetical scenarios as the community of researchers can imagine.

The paper is structured in the next sections: section 2 includes the mathematical description of the computations involved in the hydrological problem, section 3 describes the agents, roles, protocols and ontology proposed to model the interaction between all the parts. section 4 shows the implementation, GUI and parameters of the simulation tool, and finally, section 5 outlines conclusions and future work.

2. Hydraulic characterization of irrigation districts´operation

We assume individual farmers make their decisions on irrigation planning based on the soil moisture level (θ_i) estimated through a simple mass conservation law between the system inputs ($I(t)$) and outputs ($O(t)$). See eqs. 1 and 2,

$$\frac{\partial \theta}{\partial t} = I(t) - O(t) \tag{1}$$

$$\theta_i = \theta_{i-1} + In_i - ET0_i \cdot Kc_i \tag{2}$$

where:

- In_i stands for the rainfall that would infiltrate into the soil at day i, being a random variable forecasted externally by a weather station.
- $ET0_i$ stands for the evapotranspiration produced at day i, being a random variable forecasted externally by a weather station..
- Kc_i stands for the crop coefficient at day i, specific for each crop.
- θ_{i-1} stands for soil moisture in the previous day (i-1).

The soil moisture (θ_i) is physically bounded between the saturated (maximum, θ_s) and residual (minimum, (θ_r) levels. Farmers irrigate the crops if the estimated soil moisture drops below a triggering level (θ_t) seeking, ideally but dependent on the cropping strategy, to provide a volume of water to increase the soil moisture to the desired level (θ_{cc}) (eq. 3).

$$\theta_i = \begin{cases} \theta_r & \text{when} < \theta_r \\ \theta_s & \text{when} > \theta_s \\ \theta_{cc} & \text{when} < \theta_t \text{ and farmer connected to network} \end{cases} \quad (3)$$

Therefore, any farmer foreseeing the soil moisture will drop below the triggering level will demand the network to provide him with a given flow (q_i) at a desired pressure head (hd_i) over the time required to increase θ_i to θ_{cc}. The amount of farmers connected to the network will thus determine its operation conditions in terms of (real) discharged flow ($Q_r = \sum_{i=1}^{n} q_i$) and pressure head (H_r). The network capacity is given by an initial project design based on an optimal operation point (Q_p, H_p) being the real operation conditions (Q_r, H_r) often different than the optimal point what makes the farmers will be served at a real point (hr_i) different that the desired one (hd_i). Following [17], the network operation (Q_r, H_r) and the real supply conditions (hr_i) to each one of the $n \leq N$ farmers connected to the network can be then parametrized by eqs. 4, 5, 6.

$$\mathbf{Q} = \mathbf{A} \cdot \vec{q}_i \quad (4)$$

$$\vec{hfa} = \mathbf{A}^T \cdot \vec{hf}_i \quad (5)$$

$$\vec{h}_r = \vec{H}_r - \vec{hfa} \quad (6)$$

Being $\vec{q}_i$ the vector accounting for the farmers connected (q_i in case the farmer is connected, 0 otherwise), $\mathbf{A}$ the matrix defining the relationship between the network´s stretches (i, rows) and nodes (j, columns, being $A_{ij} = 1$ if water has to cross stretch i to reach the node j and 0 otherwise), $\vec{h}_r$ the vector with the real pressure head at each node (farmer), $\vec{hfa}$ the aggregated energy losses until each node and $\vec{hf}_i$ the vector with each stretch´s individual energy losses. Since $\vec{hf}_i = f(Q_i)$ (being Q_i the flow passing through the stretch i that depends on the amount of open hydrants downstream of the stretch i), the configuration of farmers connected to the network determines the conditions of the service provided to each farmers. In 6, we define $\vec{H}_r$ with N H_r values. As a reply to the system status derived from previous equations, the network delivers the energy (H_s) following eq. 7 being the performance given by eq. 8.

$$H_s = a_0 + a_1 \cdot Q_r - a_2 \cdot Q_r^2 \quad (7)$$

$$\mu = b_0 + b_1 \cdot Q_r - b_2 \cdot Q_r^2 \quad (8)$$

If the water is provided by a reservoir $H_d = z_d$ being z_d the topographical elevation of the water at the reservoir. Otherwise, when the energy is provided by a pumping station, the potential presence of frequency converters or multiple pumping systems would drive to some updates in eqs. 7 and 8.

3. Proposed Solution: Agents, ontology and protocols

Although agents can be implemented using Generative AI, its main advantage (the use of natural language communicating with human users) plays no part in our problem, while its main disadvantage (lack of explainability and controllability) plays a key role in our problem. Therefore, the proposed agents to address the problem use a cognitive approach. Among the many agent cognitive architectures that have been proposed, [9], the BDI (Beliefs, Desires and Intentions) logic is one of the most used since it combines modeling human-like reasoning [10] with a formalization in modal logic [11]. BDI agents use explicit beliefs about themselves, about their peers and about the environment jointly with explicit and understandable final and intermediate desires and plans.

We have defined our BDI system complying with the Foundation for Intelligent Physical Agents (in advance, FIPA) standard [13] inspired by the Speech Act theory [12] to ensure the expected interoperatibility of open systems.

3.1. Agents definition and communications

Based on the previous principia, to properly simulate the irrigation district operation we define three types of agents:

- Farmers (F). A set of agents deciding whether or not to ask for connection to the irrigation network to receive the flow rate q_i at head pressure hd_i over a time t to cover the crop water needs.
- Weather estimates provider (WEP). A single agent representing a weather station providing daily forecast on infiltration In and evapotranspiration $ET0$.
- Energy deliverer (ED). A single agent representing the pumping station of the network operating at (Q_r,H_r). This agent can also reject the farmers´ request.

The cycle proposed to simulate the agents operation follows these steps (see figure 1):

1. The ED triggers the process by informing the WEP on the arrival of a new date.
2. The WEP provides farmers with the weather forecasts, (In_i and $ET0_i$) for the reported date.
3. After processing the received weather forecast, each farmer F decides whether or not to request a connection to the irrigation network depending on the estimated θ_i and if positive, farmer F will send the request to the deliverer ED that replies either accepting or refusing the request.
4. Finally, the deliverer ED informs to each connected farmer F with the head pressure hr_i he is being provided with.

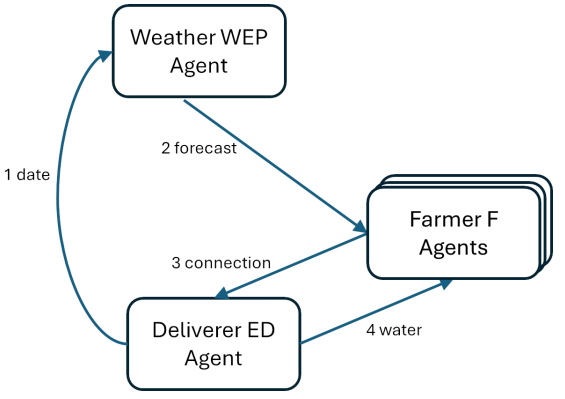


Figure 1: Information flow

The information exchanged in these communications is encapsulated in the next concepts (summarized in figure 2):

- A date concept with the single attribute when (the current date).
- A forecast concept which contains three attributes: when (the current date), the infiltration (In_i) and the evapotranspiration ($ET0_i$).
- A connection concept identifying the farmer F demanding a connection to the irrigation network, the connection decision of the farmer F and the date of such connection.
- A water concept comprising the pressure head (hr_i) the farmer F is being provided with.

The communications outlined in figure 1 are implemented through interaction protocols which are compliant with the IEEE FIPA standard [13]. Specifically, FIPA Communicative Act Library Specification (XC00037H) lists the possible performatives that each message of a FIPA-compliant protocol should have associated, which include the ones we used: query, inform, agree, refuse and request. The instances of these protocols are executed on a daily basis following the relationships listed below (see figures 3 to 6 for a graphical representation of the interactions):

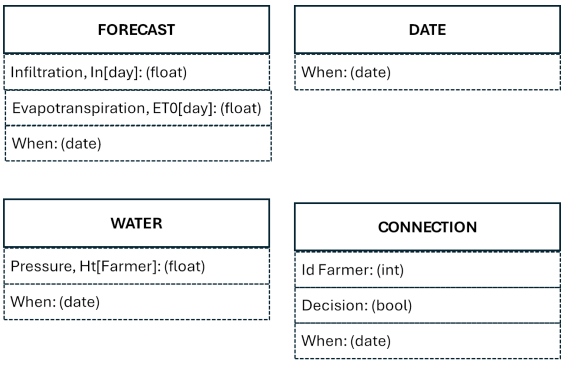


Figure 2: Concepts of the ontology

1. An initial action from the ED to both the WEP and all farmers F communicates the current date.
2. The WEP and each farmer F interact by exchanging two messages: first, one message from F asking WEP for the forecast concept and, second, a reply message from the WEP reporting on the infiltration In and the evapotranspiration $ET0$.
3. The ED and each farmer F interact by exchanging three messages: 1) each F requests a water provision through a connection to the network; 2) after evaluating ED the network capacity, individually decides and messages each farmer communicating the decision of accepting or refusing each request; 3) in case of acceptance, the ED informs the farmer F about the pressure head hr_i actually provided.
4. An alternative interaction between the ED and a farmer F occurs when the F does not need to connect to the network, informing the F to ED about its decision to keep out of the network.

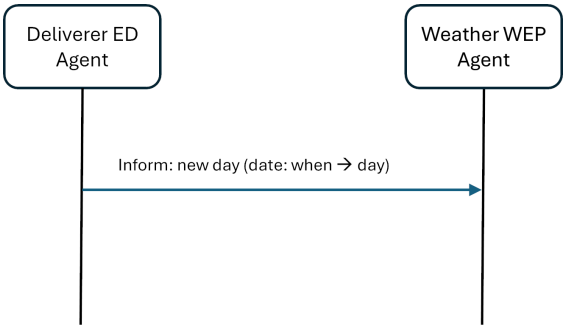


Figure 3: Informing date Protocol

3.2. Agents´ internal reasoning

Unlike ED and F that have complex internal decision making, the weather station WEP follows no BDI reasoning. This WEP agent has no memory, no plan, no goals, no internal decisions or computations. It is just a reactive agent that regularly provides the weather information demanded by farmers. This WEP agent is included in a yellow pages service called "directory facilitator" as IEEE FIPA standard suggests. This registration allows the remaining agents to identify it in the system for further communications. The WEP starts operations by gathering the infiltration and evapotranspiration data from external files and awaits for messages from the farmers to reply upon requests within both the Infiltration and evapotranspiration values of the date reported.

3.2.1. BDI reasoning of Energy deliverer

Initially, the ED is listed as facilitator following the IEEE FIPA standard thus allowing the farmers to detect it for future communications. The ED must know in advance information on the pipe network layout and dimensioning and

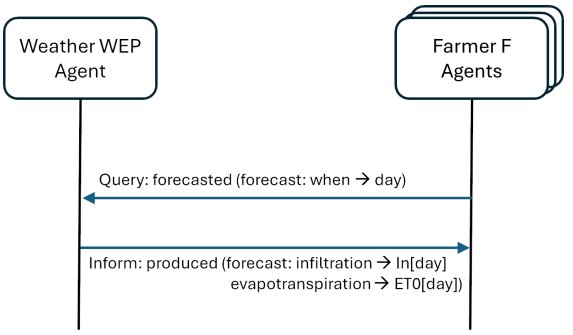


Figure 4: Querying forecast Protocol

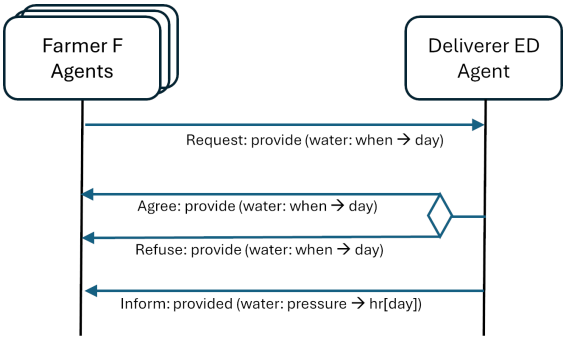


Figure 5: Requesting water Protocol

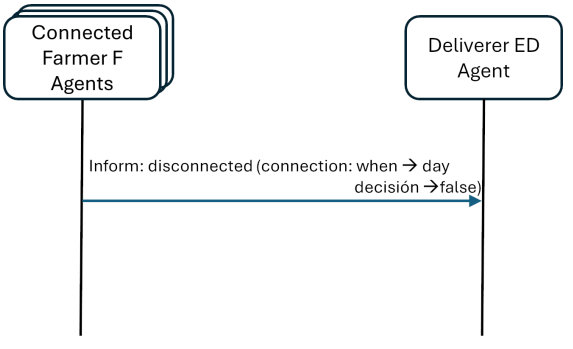


Figure 6: Informing disconnected Protocol

the remaining variables and parameters to run the hydraulic calculations presented in section 2. At the beginning, the ED takes the belief "new day" what launches the desire "spend day" thus sending messages to the farmers containing the actual day. When receiving that communication, all F ask for weather forecast, process the data to estimate soil moisture and, if required, request ED for connection. Upon any request, the ED will decide whether or not the connection is accepted. If the connection is allowed, ED adds the farmer identification, and the date, to a list of beliefs "farmer connected" and sends the corresponding acceptance message. In case the connection is not accepted, a rejection message is sent from the ED to F. Therefore the beliefs "farmer connected" of ED contains the set of farmers connected to the network in the current date.

In this process, different decision rules can be used to support :

1. Reject any connection making $\sum_{i=1}^{i=N} q_i > Q_p$ what stands for a "interventionism level" setup parameter that characterizes the ED.

2. Reject connections following external constraints as for example energy rates, pump station performance, defining some available slots or restrictions from drought conditions, etc thus defining a "max connections" setup parameter.
3. Defining allocation criteria based on some farmers characteristics linked to irrigation or water use efficiency.

At the time the ED feeds the beliefs "farmer connected", he also updates the list of beliefs "farmer updated" that triggers the desire "check all updated" to check whether or not all the farmers have updated their status. If all of them have updated the status the ED adopts the belief "all updated" and launches the desire "compute pressure" thus running the hydrological calculations presented in section 2 using the list of beliefs "farmer connected".

Once the calculations have been performed, the ED´s belief changes to "pressure computed" what triggers the desire "provide pressure" thus reporting all the farmers connected on the provided head pressure. After that, the ED´s belief changes to "new day" and starts the desire "spend day" thus re-initializing the process. The whole reasoning process is illustrated in figure 7 where messages are colored in red and beliefs are colored in blue.

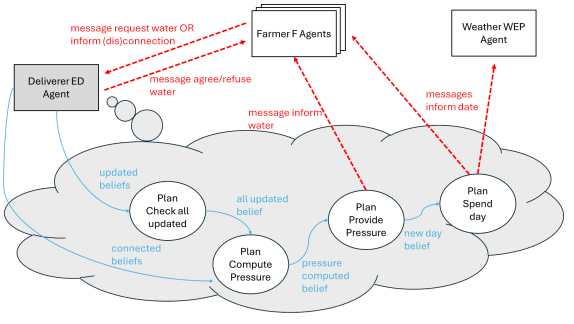


Figure 7: BDI reasoning of the deliverer ED agent

3.2.2. Reasoning of farmer agents

Farmers are defined by their location within the network and the expected flow rate (q_i) and pressure head (hd_i) values. The previous information can be gathered from external graphml of GIS files. Also any farmer must have an individual identification and information on the crop name, crop coefficients

The first actions they take is finding ED and WEP in the directory facilitator and adopting the initial beliefs "weather unknown" and "disconnected". The first belief launches the desire "forecast" that implies asking WEP for the weather forecast. After receiving the message from WEP, farmer F computes the soil moisture (see eq. 2). In case the soil moisture estimate θ_i drops below the triggering level θ_t, F will adopt the belief "water required", being "water not required" otherwise. The previous situation drives to several beliefs combinations depending on both F and ED´s status as defined below:

- A combination of beliefs "water required" and "disconnected" launches the desire "connect""
- A combination of beliefs "water required" and "enough pressure" launches the desire "connect". The belief "enough pressure" would be adopted by the farmer when, while connected, the received pressure head satisfies at least a minimum threshold.
- A combination of beliefs "water required" and not "enough pressure" launches the desire "rethink connection".
- A combination of beliefs "water not required" and "connected" launches the desire "disconnect"

Once the farmers are connected, they can behave differently and some different criteria can be incorporated to decide when they disconnect:

1. They have received the volume of water they estimate can cover the crop requirements for the pre-defined period of time.
2. A previously defined time-lapse between irrigation applications has been completed..
3. They have received the volume of water to increase θ_i to θ_{cc} or a fraction of that value.

4. Any other rational or irrational farmers´ behavior based on subjective perception and incomplete information in relation to the remaining farmers.

The connected farmers also evaluate the need to keep in the network by adopting the believe "rethink" to decide whether (desire "connected") or not (desire "disconnected") they remain connected. Once the farmers have received the demanded water with the corresponding head pressure, they will then adopt the status "disconnect". It means reporting the ED on the disconnection and adopting the beliefs "disconnected" and "weather unknown" to re-initialize the process.

The whole reasoning process is illustrated in figure 8 where messages are colored in red and beliefs are colored in blue.

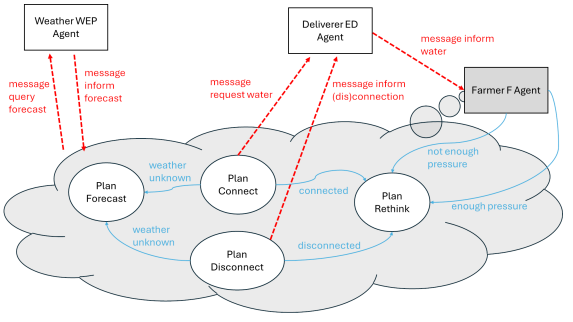


Figure 8: BDI reasoning of a farmer F agent

4. Implementation, parameters and GUI of the simulator

Nowadays, a great number of agent platforms and languages for developing agent systems can be found in the literature. The GAMA agent platform [14] is perfectly suited for simulating irrigation districts, since it is, on the one hand, an open source software which is FIPA compliant and includes the possibility of using the BDI paradigm in the reasoning [15] and, on the other hand, it allows large scale simulations integrating graphml and GIS data [16].

4.1. Model parametrization and initial setup

We have used the synthetic irrigation network presented in figure 9 previously used as case study in [17]. In this

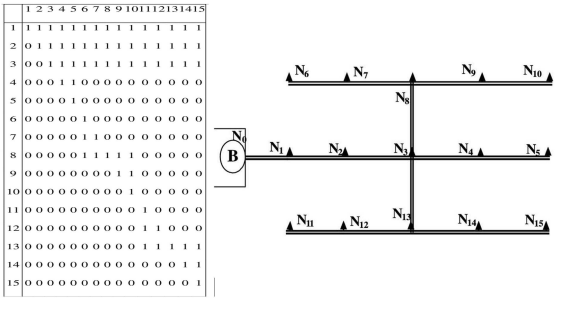


Figure 9: Network layout

case the GAMA platform was fed on the graphml file being the platform parametrized with the information presented in tables 1 and 4 directly gathered from the graphml file.

The following input files are also required in this model´s configuration:

- A weather file that contains daily values of infiltration and evapotranspiration over a 1 year period.
- A crop file that containing each farmer´s cultivated crop (farmer id, planted crop).

Table 1
Map file (GIS or Graphml) crossing attributes used in the GAMA simulation

Name	Meaning	datatype
id	Farmer´s identification code	string
q	Desired flow rate (q_i)	float (m^3)
height	Topographical altitude of the farmer	float (m)

Table 2
Required beliefs and concluding desire of farmer agent

Input Beliefs	Output Desire
weather unknown	forecast
required AND disconnected	connect
required AND pressure	connect
required AND not pressure	rethink
required AND not pressure	rethink
not required	disconnect

Table 3
Required beliefs and concluding desire of deliverer agent

Input Beliefs	Output Desire
new update	check all updated
all updated	compute pressure
new day	spend day
water computed	provide water

Table 4
Map file (GIS or Graphml) channel attributes used in the GAMA simulation

Name	Meaning	datatype
source	Initial node	string
target	Final node	string
length	Pipe length	float (m)

- As many files as potential cultivebale crops with the daily crop coefficients (Kc_i) over a year.

We have supposed the following values for parametrizing the network $hd_i = 20m$, all pipes´ diameters $D = 0.26m$, $Q_p = 0.08m^3/s$ and $H_p = 31.06m$. Similarly, regarding the farmers soil moisture estimates, we have supposed $\theta_s = 0.42$, $\theta_r = 0.078$, $\theta_{cc} = 0.3$ and $\theta_t = 0.2$. The previous parameters can be easily tuned by directly introducing the correct values over the GUI (see figure 10 where a GUI´s screen capture is presented).

The resulting agent system runs over the network along the number of days included in the input txt file for the effective rainfall and evapotranspiration. Figures 11 and 12 present some images of the initial network setup and the network state at a given time where the pumping station is colored in blue, the farmers watering green colored while those not connected to the network are represented in red color. Pipes with positive velocity are depicted in green color and pipes with zero velocity are black. The nodes´ size has been defined to be countiously updated proportional to the pressure head received in that connecting pipe. Once the simulation starts, the colors of nodes and sizes of arrows dynamically change showing the evolution of the water distribution along the corresponding days.

The resulting agent system runs over the defined time period number showing in the output file and in the log window the corresponding information about the steps of BDI, FIPA and computation steps taken by each agent. For

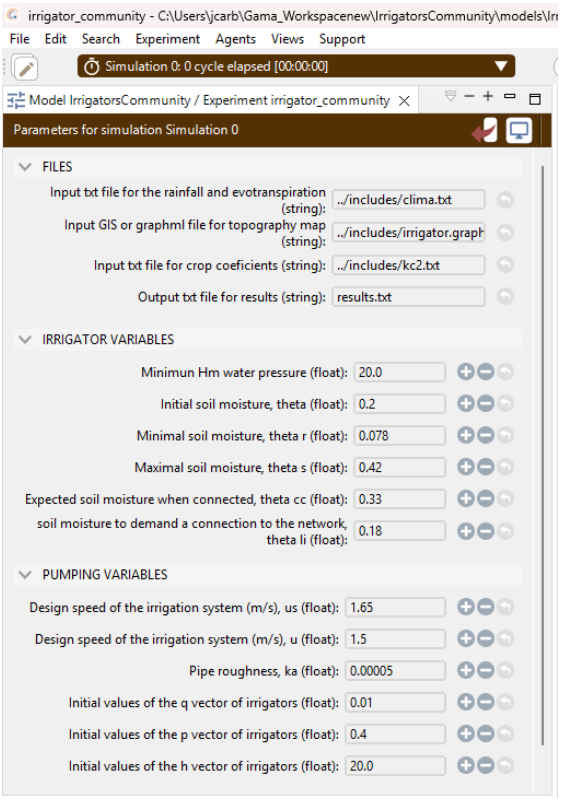


Figure 10: GUI´s screen capture with the setup parameters

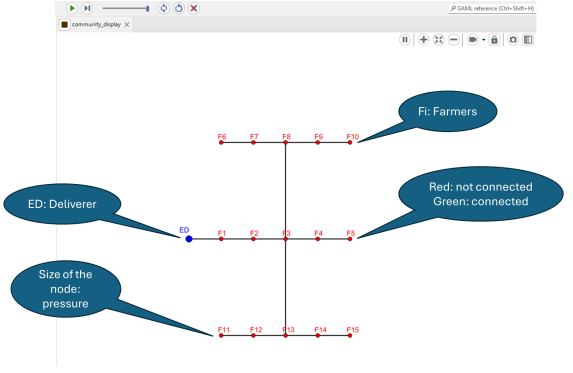


Figure 11: GUI´s screen capture with the initial state of the proposed illustrative network

instance, let us take a particular farmer and the deliverer to show a case step by step. First, the initial beliefs of the farmer are adopted:

```
Farmer 6  BDI : adopted belief crop of Onion
Farmer 6  BDI : adopted belief of minimal soil moisture of 0.18
Farmer 6  BDI: adopting belief disconnected
Farmer 6  BDI: adopting belief weather unknown
```

Once farmer 6 adopts the belief "weather unknown", it will then adopt and implement the desire forecast:

```
Farmer 6  BDI: adopted desire forecast
```

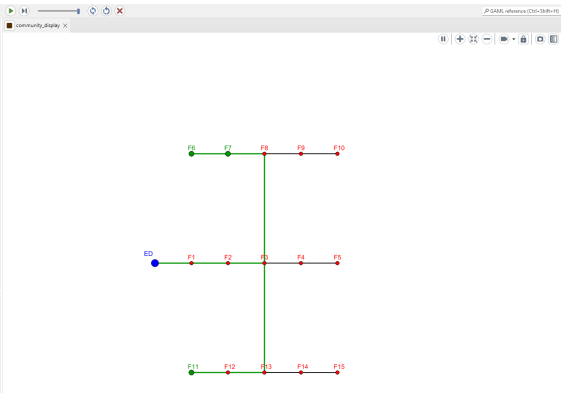


Figure 12: GUI´s screen capture with an intermediate state of the proposed illustrative network

```
Farmer 6  BDI: dropping belief weather unknown
Farmer 6  FIPA: query forecast message sent to Weather
Weather FIPA: agree forecast message sent to farmer 6
Farmer 6  BDI: dropping desire forecast
```

With the infiltration and evapotranspiration of the current day received in the forecast message, the farmer computes whether it requires water or not, therefore adopting alternatively the desire water required or not water required beliefs:

```
Farmer 6  computation: current theta is 0.15
Farmer 6  computation: there is not enough soil moisture
Farmer 6  BDI: adopting water required belief
Farmer 6  BDI: adopted desire connect
Farmer 6  FIPA: request water message sent to Deliverer
Farmer 6  BDI: dropping water required belief
Farmer 6  BDI: dropping desire connect
Deliverer BDI: adopting updated belief of farmer 6
Deliverer BDI: adopting new update belief
```

Next, the deliverer, when all the farmers have communicated whether or not demand connection, will accept or reject the connection requests:

```
Deliverer BDI: adopted desire check all updated
Deliverer BDI: dropping new update belief
Deliverer computation: all farmers already updated
Deliverer FIPA: agree water message sent to farmer 6
Farmer 6  BDI: dropping disconnected belief
Deliverer BDI: adopting connected belief of farmer 6
Deliverer BDI: adopting all updated belief
Deliverer BDI: dropping desire check all updated
```

Afterwards, the deliverer proceeds to compute the pressure that all connected farmers would receive:

```
Deliverer BDI: adopted desire compute pressure
Deliverer computation: ht of the farmer 6  is 28.21860864570708
Deliverer computation: ht of the farmer 11  is 28.21860864570708
Deliverer computation: Qp is 0.07980000000000002
Deliverer computation: SumQ is 0.45000000000000023
Deliverer BDI: adopting water computed belief
Deliverer BDI: dropping desire compute pressure
```

And finally, the deliverer reports connected farmers on the provided pressure head while each farmer evaluates whether such pressure is enough or not:

```
Deliverer BDI: adopted desire provide pressure
Deliverer BDI: dropping water computed belief
Deliverer BDI: adopting new day 0 belief
Deliverer BDI: dropping desire provide pressure
Farmer 6 computation: pressure received is greater than the minimum threshold
Farmer 6  BDI: adopting enough pressure belief
Farmer 6  BDI: adopting connected belief
```

The output file includes the following information:

- Days of each farmer being rejected for connection.
- Days of each farmer without enough soil moisture.
- Days of each farmer without enough head pressure.
- Head losses and energy head at each farmer´s hydrant.
- Farmers´estimated soil moisture.

4.2. Future developments, potential application in real case studies

The developed code is also suited for simulating real complex irrigation networks. The model can be fed on GIS files with the network´s layout and hydrants, pipes and energy deliverer characteristics. In figure 13 we present the GIS layout from the "Comunidad de Regantes de la Margen Izquierda del Porma" irrigation district in León (Spain) [1] that has been imported to the developed agent simulation architecture while figure 14 shows the model parametrization with this real case study.

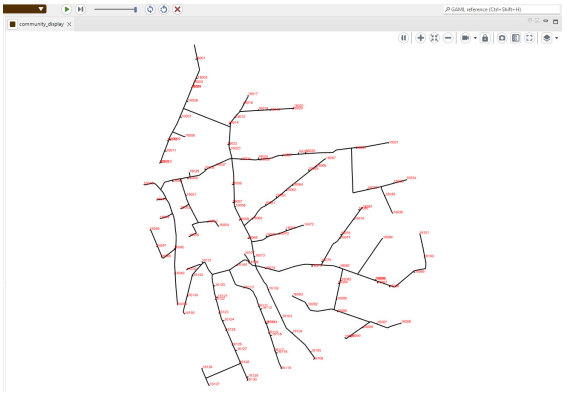


Figure 13: GIS Network imported by our agent simulator

A number of different functionalities can be added to the basic setup we have presented in this manuscript. The model is perfectly scalable and can be easily tuned for incorporating enhanced modeling tools. The code underlying the agents definition has been implemented by Python function to help use Specific functions can be easily added to consider the following items:

- The energy deliverer agent can be improved by incorporating complex pumping stations operation including groups of pumping units, running in parallel, bounding the provided pressure head between target levels or pump frequency converters. Specific functions to manage connection requests according to energy costs, farmers´ water use efficiency, time-lapse between lasts irrigation, penalties to farmers requesting connection and later neglecting when among many others can be implemented to

[1] https://www.crporma.com/, last visit June 24th 2025

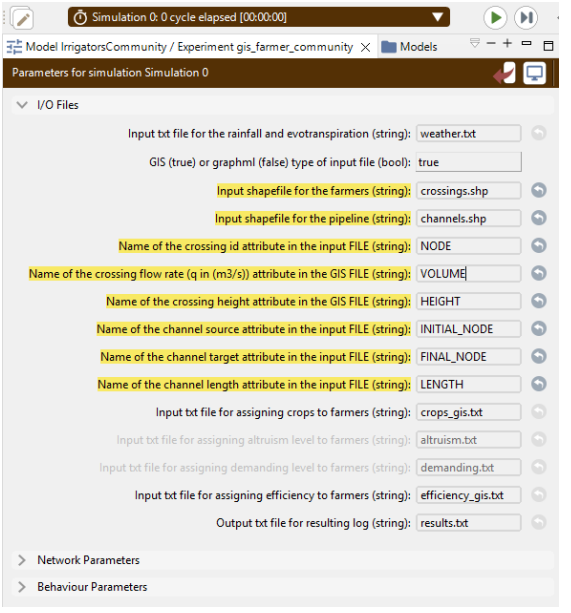


Figure 14: File parameters of the simulator in GAMA

- The weather estimates provider can be incorporate more complex weather forecasting functions or spatial functionalities as for example those from drones, satellite information or spatially distributed sensor networks to deliver more accurate estimates.

- The farmers operation conditions can be improved by incorporating individual criteria or traditional behavior as egoism/altruism or cooperative actions seeking to obtain a better supply from the network (see figure 15 where an initial version incorporating enhanced farmers´behavior has been drafted). Similarly the faramers´ decision on cropping strategy can be largely improved by incorporating particularly tuned functions for modeling crop evolution, soil water dynamics, etc.

In particular, we suggest some future works would address the use of this simulator to:

- obtain a comparison of our distributed approach with a centralized complete knowledge irrigation system where the deliverer agent decides on behalf of all the farmers.

- measure the damage caused by different percentages of each type of agents (egoists, demanding farmers, and interventionist deliverer) present in the system.

- study the improved efficiency that direct peer to peer negotiation (reaching quid pro quo agreements) between farmers could achieve.

- analyze the possibility of using an Electronic Institution [18] to monitor and to enforce good behavior of farmers through the deterrence of potential economic or legal sanctions applied by the electronic institution itself.

- include UAV agents that would provide relevant information on the crop status or weather variables to farmer agents through a cooperative permanent and cycling surveillance of irrigation zones as [19].

5. Conclusions

This paper has explored the viability of a distributed alternative way of representing an irrigation network including both the computations of a hydrological model and the farmers´motivated individual decision making. To the best of our knowledge, this has not been addressed before in the scientific literature. We have succesfully merged both domains by

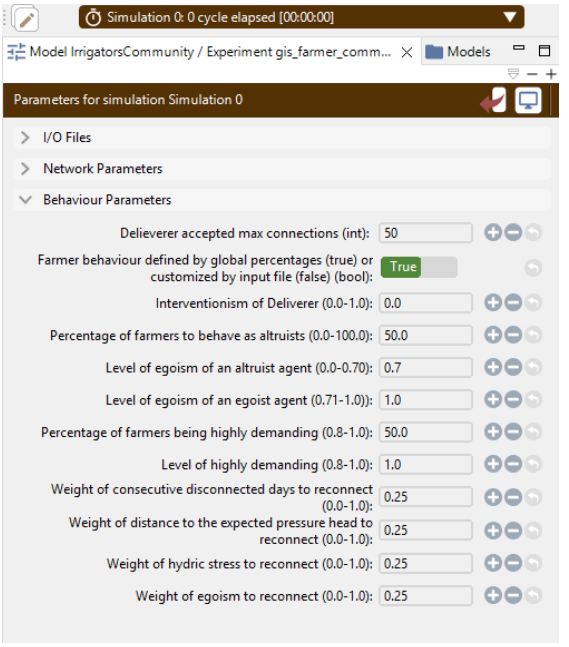


Figure 15: Farmers´ behaviour parameters of the simulator in GAMA

- Describing the associated computations aimed at estimating the provided pressure head to each farmer connected to the network.
- By defining specific agents to represent farmers operation we leverage the distributed technology provided by agents theory to properly simulate the farmers´s decision preserving their privacy and autonomy to farmers.
- We have suggested a set of roles, an ontology of actions, predicates and concepts and the protocols to model the flow of information exchanged between the agents in way compliant with the IEEE standard named FIPA.
- We have explicitly characterized the underlying logic of decisions taken by deliverer and farmer agents in order to achieve controllability and explainability. Such logic is expressed through the BDI reasoning of the deliverer agent and the farmer agents which allows the explicit description of different sequential reasoning steps that could take place.
- We have implemented in GAML (the programming language of GAMA agent platform) this model into a flexible simulator, easy to parametrize, which was uploaded to a public repository.

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Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study. The complete code of all the agents jointly with the environment and setup files are publicly available in the Sourceforge repository in order to provide transparency, to promote further use by the research community and facilitate the complete replicability of all the simulations included in this manuscript.

Declaration of generative AI and AI-assisted technologies in the writing process: During the preparation of this work the author(s) did not use generative AI nor AI-assisted technologies in the writing process.

The complete code of all the agents jointly with the environment and setup files for both, Graphml and GIS topologies are publicly available in the Sourceforge project named "irrigators network" [2] in order to provide transparency and facilitate the complete replicability of all computations included in this manuscript.

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[2] https://sourceforge.net/projects/irrigatorsnetwork/



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